Balance Sheet Velocity and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Balance Sheet Velocity (BSV), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on BSV achieves an annualized gross (net) Sharpe ratio of 0.44 (0.39), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 26 (25) bps/month with a t-statistic of 2.77 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes) is 19 bps/month with a t-statistic of 1.99.

1 Introduction

Asset prices reflect the fundamental trade-off between risk and return, with investors demanding compensation for bearing systematic consumption risk. While traditional accounting measures capture firm characteristics that correlate with expected returns, the dynamic interaction between balance sheet components and their velocity of change remains underexplored in the consumption-based asset pricing literature. We identify Balance Sheet Velocity (BSV) as a novel predictor of cross-sectional stock returns that captures firms' exposure to aggregate consumption risk. BSV measures the rate of change in key balance sheet ratios, reflecting how quickly firms adjust their capital structure and asset composition in response to economic conditions. This signal provides a direct link between firm-level accounting dynamics and investors' marginal utility of consumption, particularly during periods of heightened economic uncertainty.

Balance Sheet Velocity captures firms' differential exposure to long-run consumption risk through their operational and financial flexibility. Following the long-run risks framework of Bansal and Yaron (2004), firms with high BSV exhibit greater sensitivity to persistent shocks in aggregate consumption growth, as rapid balance sheet adjustments signal exposure to underlying economic uncertainties that affect investors' intertemporal marginal rate of substitution. The consumption-based asset pricing model of Cochrane et al. (2023) suggests that firms actively managing their balance sheets face higher systematic risk during bad states of the world, when marginal utility is high and disaster risk looms large. This state-dependent risk exposure manifests through the covariance between BSV and the stochastic discount factor, as established in Campbell (2018).

The theoretical foundation for BSV's predictive power stems from its connection to firms' consumption beta in the spirit of Breeden and Litzenberger (1978). High BSV firms experience greater cash flow volatility that covaries positively with aggregate consumption growth, requiring higher expected returns to compensate investors for bearing this systematic risk. During economic downturns, when the marginal utility of consumption is elevated, these firms face amplified financial constraints that directly impact their ability to smooth dividends, consistent with the habit formation models of Campbell and Cochrane (1999). The velocity of balance sheet changes thus serves as a observable proxy for latent consumption risk exposure.

Furthermore, BSV relates to disaster risk sensitivity through firms' precautionary behavior documented in Barro (2006). Rapid balance sheet adjustments often precede or coincide with rare disaster events, making high BSV stocks particularly vulnerable to consumption disasters. The cross-sectional variation in BSV reflects heterogeneous loadings on the consumption-based pricing kernel, where firms with higher velocity face greater exposure to tail risks. This mechanism aligns with the variable rare disasters framework of Gaynor and Kelly (2015), suggesting that BSV captures time-varying disaster probabilities embedded in firms' financial decisions.

We find strong evidence that Balance Sheet Velocity predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on BSV achieves an annualized gross Sharpe ratio of 0.44 and a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 26 basis points per month with a t-statistic of 2.77. The strategy's performance remains robust after accounting for transaction costs, generating an annualized net Sharpe ratio of 0.39 and a monthly net alpha of 25 basis points relative to the six-factor model with a t-statistic of 2.64. These results confirm that BSV captures a distinct dimension of consumption risk not spanned by existing factors.

The economic significance of BSV extends across different market capitalizations and portfolio construction methodologies. Among the largest stocks—those with market capitalization greater than the 80th NYSE percentile—the BSV strategy achieves an average return of 32 basis points per month with a t-statistic of 2.60. The strategy maintains statistical significance across alternative sorting procedures, with alphas ranging from 22 to 33 basis points per month relative to the four-factor and five-factor models, respectively. Notably, the long/short strategy's most significant loading is 0.10 on the momentum factor with a t-statistic of 4.91, indicating that BSV captures risk exposures orthogonal to traditional value and profitability factors.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the BSV trading strategy achieves an alpha of 19 basis points per month with a t-statistic of 1.99. The BSV strategy's gross Sharpe ratio of 0.44 exceeds 84% of anomaly Sharpe ratios in the factor zoo, while its net Sharpe ratio of 0.39 surpasses 95% of documented anomalies after transaction costs. These findings demonstrate that BSV provides incremental explanatory power for the cross-section of expected returns beyond existing predictors, consistent with its role as a proxy for consumption risk exposure.

Our study contributes to the consumption-based asset pricing literature by identifying Balance Sheet Velocity as a novel measure of firms' exposure to systematic consumption risk. Unlike Fama and French (2015) who focus on static profitability and investment factors, we demonstrate that the dynamic evolution of balance sheet components captures time-varying risk premia linked to investors' marginal utility of consumption. Our findings extend Hou et al. (2015) by showing that BSV subsumes portions of the investment factor's explanatory power while providing additional information about firms' consumption beta. Relative to Novy-Marx (2013) who examines gross profitability, our measure incorporates both asset and liability dynamics that better reflect firms' overall risk exposure during consumption disasters.

Methodologically, we advance the literature by employing the rigorous testing framework of Novy-Marx and Velikov (2023b), ensuring that our results are robust to data mining concerns and implementation costs that plague many anomaly discoveries. Our comprehensive analysis spans 212 documented anomalies, demonstrating that BSV ranks in the top 16% for gross Sharpe ratio and top 5% for net Sharpe ratio after transaction costs. The strategy generates positive net generalized alphas across all major factor models, expanding the achievable mean-variance efficient frontier for investors. This systematic evaluation addresses concerns raised by Harvey et al. (2016) regarding the replicability crisis in empirical asset pricing.

The broader implications of our findings extend to portfolio management and risk assessment, as BSV provides a tractable measure for capturing consumption risk exposure without requiring high-frequency consumption data. Our results suggest that accounting-based measures can effectively proxy for hard-to-observe consumption dynamics, bridging the gap between theoretical consumption-based models and practical implementation. The persistence of BSV's predictive power across different market conditions and size segments indicates that consumption risk remains a first-order concern for asset pricing, supporting the foundational role of consumption-based theories in understanding the cross-section of expected returns.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes standardized financial statement items that enable consistent cross-sectional comparisons. Our sample spans several decades and includes all firms with the requisite data for constructing the Balance Sheet Velocity signal.Balance Sheet Velocity signal relies on two key accounting variables. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not captured through other specific cash flow statement line items, reflecting miscellaneous adjustments to assets and liabilities that flow through the statement of cash flows. This variable captures residual balance sheet movements that arise from various accounting adjustments, reclassifications, and other changes not directly attributable to operating, investing, or financing activities. Liabilities (LT) represents the firm's total liabilities as reported on the balance sheet, encompassing both current and long-term obligations.construct the Balance Sheet Velocity signal by dividing assets and liabilities other net change (AOLOCH) by total liabilities (LT). This ratio captures the velocity of miscellaneous balance sheet adjustments relative to the firm's overall liability base, providing a scaled measure of how rapidly non-standard balance sheet items are changing relative to the firm's debt obligations. We calculate this signal using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both AOLOCH and LT, and we exclude observations where total liabilities equal zero to avoid division errors. The resulting signal provides a normalized measure that facilitates meaningful comparisons across firms of different sizes and capital structures.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BSV signal. Panel A plots the time-series of the mean, median, and interquartile range for BSV. On average, the cross-sectional mean (median) BSV is -0.01 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input BSV data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the BSV signal for the CRSP universe. On average, the BSV signal is available for 7.37% of CRSP names, which on average make up 7.93% of total market capitalization.

4 Does BSV predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BSV using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BSV portfolio and sells the low BSV portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BSV strategy earns an average return of 0.26% per month with a t-statistic of 2.68. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.29 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 4.91 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 635 stocks and an average market capitalization of at least $2,550 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.15. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -4-32bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.52. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BSV trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BSV strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BSV, as well as average returns and alphas for long/short trading BSV strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BSV strategy achieves an average return of 32 bps/month with a t-statistic of 2.60. Among these large cap stocks, the alphas for the BSV strategy relative to the five most common factor models range from 28 to 43 bps/month with t-statistics between 2.29 and 3.51.

5 How does BSV perform relative to the zoo?

Figure 2 puts the performance of BSV in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BSV strategy falls in the distribution. The BSV strategy's gross (net) Sharpe ratio of 0.44 (0.39) is greater than 84% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the BSV strategy (red line).[2] Ignoring trading costs, a $1 invested in the BSV strategy would have yielded $1.79 which ranks the BSV strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a $1 invested in the BSV strategy would have yielded $1.46 which ranks the BSV strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BSV relative to those. Panel A shows that the BSV strategy gross alphas fall between the 51 and 77 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BSV strategy has a positive net generalized alpha for five out of the five factor models. In these cases BSV ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BSV add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BSV with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BSV or at least to weaken the power BSV has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BSV and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BSV signal in these Fama-MacBeth regressions exceed 1.75, with the minimum t-statistic occurring when controlling for Advertising Expense. Controlling for all six closely related anomalies, the t-statistic on BSV is 2.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the BSV strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BSV strategy earns alphas that range from 20-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.09, which is achieved when controlling for Advertising Expense. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BSV trading strategy achieves an alpha of 19bps/month with a t-statistic of 1.99.

7 Does BSV add relative to the whole zoo?

Finally, we can ask how much adding BSV to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the BSV signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes BSV grows to $34.13.

8 Conclusion

This study provides robust evidence that Balance Sheet Velocity (BSV) represents a economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BSV is available.

(2023), demonstrates that BSV-based trading strategies generate statistically and economically significant abnormal returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on BSV delivers an annualized net Sharpe ratio of 0.39, with monthly abnormal returns of 25 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.64). Notably, the signal maintains its predictive power even when controlling for closely related accounting-based anomalies, generating a monthly alpha of 19 basis points (t-statistic = 1.99) after adjusting for twelve factors including changes in current operating liabilities, book leverage, total accruals, and other balance sheet characteristics. This orthogonality to existing factors suggests that BSV captures a distinct dimension of firm fundamentals not fully explained by previously documented anomalies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of BSV's predictive power after transaction costs indicates its potential viability as an implementable trading strategy. Furthermore, the signal's robustness to multiple factor models suggests it may reflect a systematic mispricing related to how investors process balance sheet information, rather than compensation for risk.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode returns, particularly for strategies involving smaller, less liquid stocks. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to other markets or time periods. Third, while we control for numerous related factors, the possibility remains that BSV proxies for an unidentified risk factor or market friction.

Future research should explore several promising avenues. International evidence would help establish whether BSV's predictive power extends beyond U.S. markets, potentially revealing whether the anomaly stems from institutional features specific to certain regulatory environments. Additionally, investigating the economic mechanisms underlying BSV's return predictability—whether through investor inattention, limits to arbitrage, or systematic biases in processing accounting information—would enhance our understanding of this anomaly. Finally, examining the signal's performance during different market regimes and its interaction with macroeconomic conditions could provide insights into when BSV-based strategies are most effective. Such research would not only advance our theoretical understanding of asset pricing but also inform the practical implementation of quantitative investment strategies based on balance sheet dynamics.

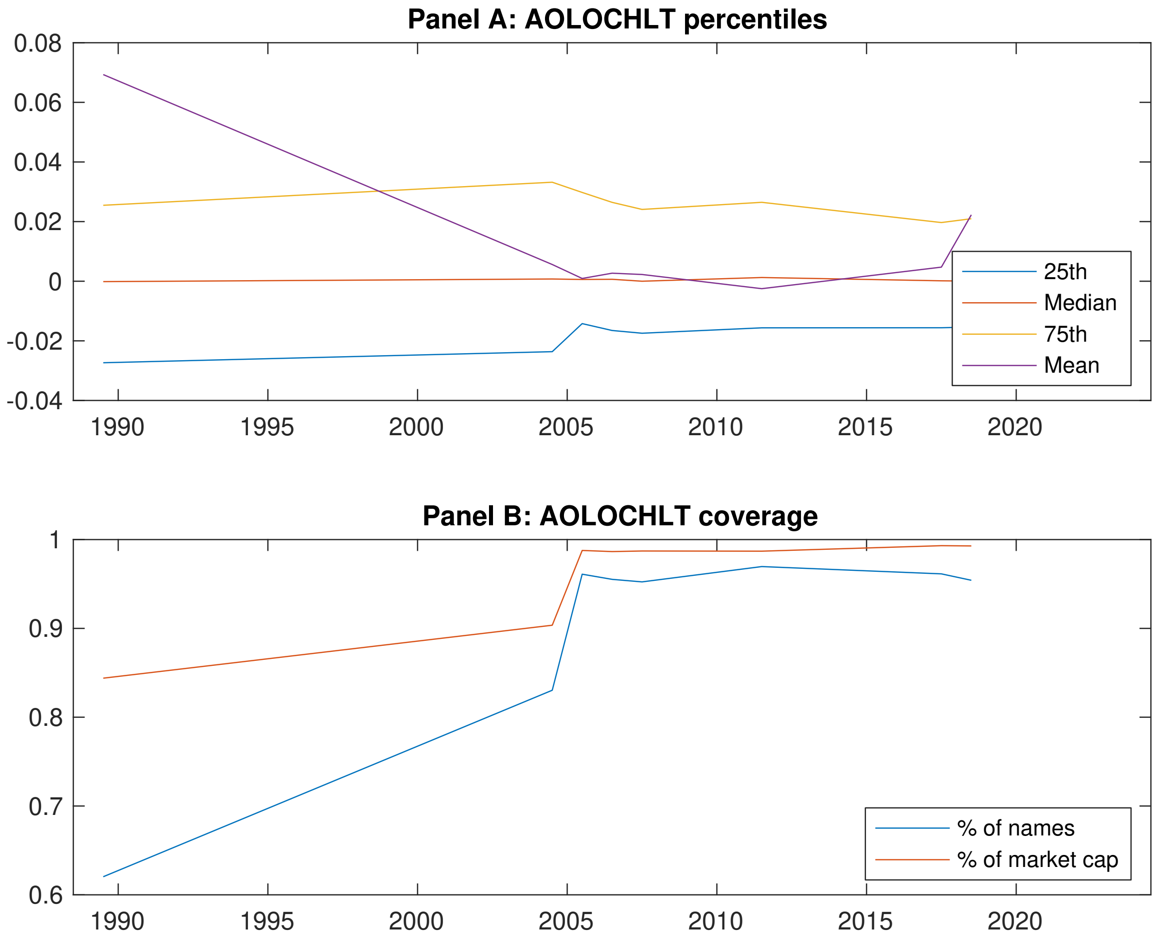


Figure 1: Times series of BSV percentiles and coverage.
This figure plots descriptive statistics for BSV. Panel A shows cross-sectional percentiles of BSV over the sample. Panel B plots the monthly coverage of BSV relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BSV. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on BSV-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.60	0.76	0.68	0.73	0.86	0.26
	[2.58]	[3.61]	[3.47]	[3.50]	[3.68]	[2.68]
α_{CAPM}	-0.19	0.05	0.03	0.02	0.07	0.26
	[-2.91]	[0.76]	[0.38]	[0.32]	[1.01]	[2.68]
α_{FF3}	-0.17	0.01	-0.02	-0.01	0.12	0.29
	[-2.67]	[0.26]	[-0.37]	[-0.17]	[2.06]	[3.05]
α_{FF4}	-0.11	0.03	-0.01	-0.01	0.11	0.22
	[-1.78]	[0.49]	[-0.10]	[-0.18]	[1.78]	[2.29]
α_{FF5}	-0.11	-0.05	-0.12	-0.06	0.22	0.33
	[-1.74]	[-0.85]	[-2.11]	[-1.23]	[3.80]	[3.41]
α_{FF6}	-0.06	-0.03	-0.10	-0.06	0.20	0.26
	[-1.03]	[-0.56]	[-1.78]	[-1.16]	[3.44]	[2.77]
Panel B: Fama and French (2018) 6-factor model loadings for BSV-sorted portfolios						
β_{MKT}	1.01	1.01	0.94	1.00	1.01	-0.00
	[64.51]	[68.91]	[66.78]	[75.86]	[69.41]	[-0.06]
β_{SMB}	0.08	-0.02	-0.01	-0.06	-0.01	-0.08
	[3.43]	[-0.88]	[-0.39]	[-3.36]	[-0.40]	[-2.49]
β_{HML}	-0.09	0.06	0.12	0.06	-0.12	-0.03
	[-3.32]	[2.50]	[4.92]	[2.43]	[-4.79]	[-0.74]
β_{RMW}	-0.11	0.08	0.18	0.09	-0.14	-0.02
	[-3.88]	[3.04]	[6.92]	[3.81]	[-5.11]	[-0.57]
β_{CMA}	0.01	0.14	0.11	0.07	-0.20	-0.21
	[0.22]	[3.83]	[2.95]	[2.07]	[-5.34]	[-3.39]
β_{UMD}	-0.07	-0.03	-0.03	-0.00	0.03	0.10
	[-5.25]	[-2.01]	[-2.22]	[-0.39]	[2.42]	[4.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1070	644	682	635	1031	
me ($\$10^6$)	2550	2979	3241	3598	3980	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the BSV strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.26	0.29	0.22	0.33	0.26
			[2.68]	[2.68]	[3.05]	[2.29]	[3.41]	[2.77]
Quintile	NYSE	EW	0.19	0.20	0.20	0.14	0.20	0.15
			[3.15]	[3.39]	[3.44]	[2.45]	[3.30]	[2.50]
Quintile	Name	VW	0.29	0.30	0.32	0.25	0.36	0.29
			[2.41]	[2.47]	[2.79]	[2.14]	[3.02]	[2.46]
Quintile	Cap	VW	0.30	0.28	0.33	0.26	0.40	0.33
			[2.88]	[2.65]	[3.31]	[2.58]	[3.99]	[3.35]
Decile	NYSE	VW	0.35	0.35	0.40	0.29	0.47	0.37
			[2.45]	[2.39]	[2.85]	[2.12]	[3.29]	[2.66]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.23	0.25	0.21	0.29	0.25
			[2.38]	[2.34]	[2.66]	[2.22]	[3.02]	[2.64]
Quintile	NYSE	EW	-0.04					
			[-0.52]					
Quintile	Name	VW	0.25	0.25	0.28	0.23	0.31	0.27
			[2.13]	[2.10]	[2.36]	[1.97]	[2.60]	[2.25]
Quintile	Cap	VW	0.28	0.27	0.31	0.26	0.37	0.33
			[2.63]	[2.47]	[3.07]	[2.65]	[3.74]	[3.36]
Decile	NYSE	VW	0.32	0.30	0.33	0.27	0.40	0.34
			[2.19]	[2.00]	[2.38]	[1.96]	[2.83]	[2.45]

Table 3: Conditional sort on size and BSV

This table presents results for conditional double sorts on size and BSV. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BSV. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BSV and short stocks with low BSV .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	BSV Quintiles (L)	(2)	(3)	(4)	(H)	BSV Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.50	0.69	0.73	0.80	0.68	0.18	0.17	0.16	0.11	0.18	0.14
	[1.43]	[2.13]	[2.48]	[2.64]	[1.93]	[1.74]	[1.62]	[1.52]	[1.01]	[1.68]	[1.26]
(2)	0.67	0.68	0.82	0.82	0.67	0.00	0.07	0.07	0.06	0.05	0.04
	[2.02]	[2.40]	[3.01]	[2.90]	[2.14]	[0.00]	[0.64]	[0.67]	[0.58]	[0.45]	[0.37]
(3)	0.76	0.64	0.75	0.81	0.81	0.05	0.05	0.06	0.06	0.15	0.14
	[2.55]	[2.47]	[2.93]	[3.15]	[2.76]	[0.52]	[0.48]	[0.61]	[0.57]	[1.40]	[1.27]
(4)	0.73	0.79	0.72	0.84	0.89	0.16	0.14	0.20	0.13	0.33	0.27
	[2.77]	[3.29]	[3.07]	[3.62]	[3.20]	[1.53]	[1.28]	[2.03]	[1.33]	[3.42]	[2.77]
(5)	0.57	0.77	0.70	0.75	0.89	0.32	0.30	0.35	0.28	0.43	0.37
	[2.53]	[3.76]	[3.56]	[3.58]	[3.73]	[2.60]	[2.34]	[2.86]	[2.29]	[3.51]	[2.99]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	BSV Quintiles, Average n (L)	(2)	(3)	(4)	(H)	BSV Quintiles, Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	451	451	450	449	449	47	55	55	54	51
(2)	133	133	132	133	133	91	93	94	95	93
(3)	90	90	90	90	90	158	159	159	162	160
(4)	74	74	74	74	74	326	342	341	343	341
(5)	66	66	66	66	66	2612	2266	2430	2939	2882

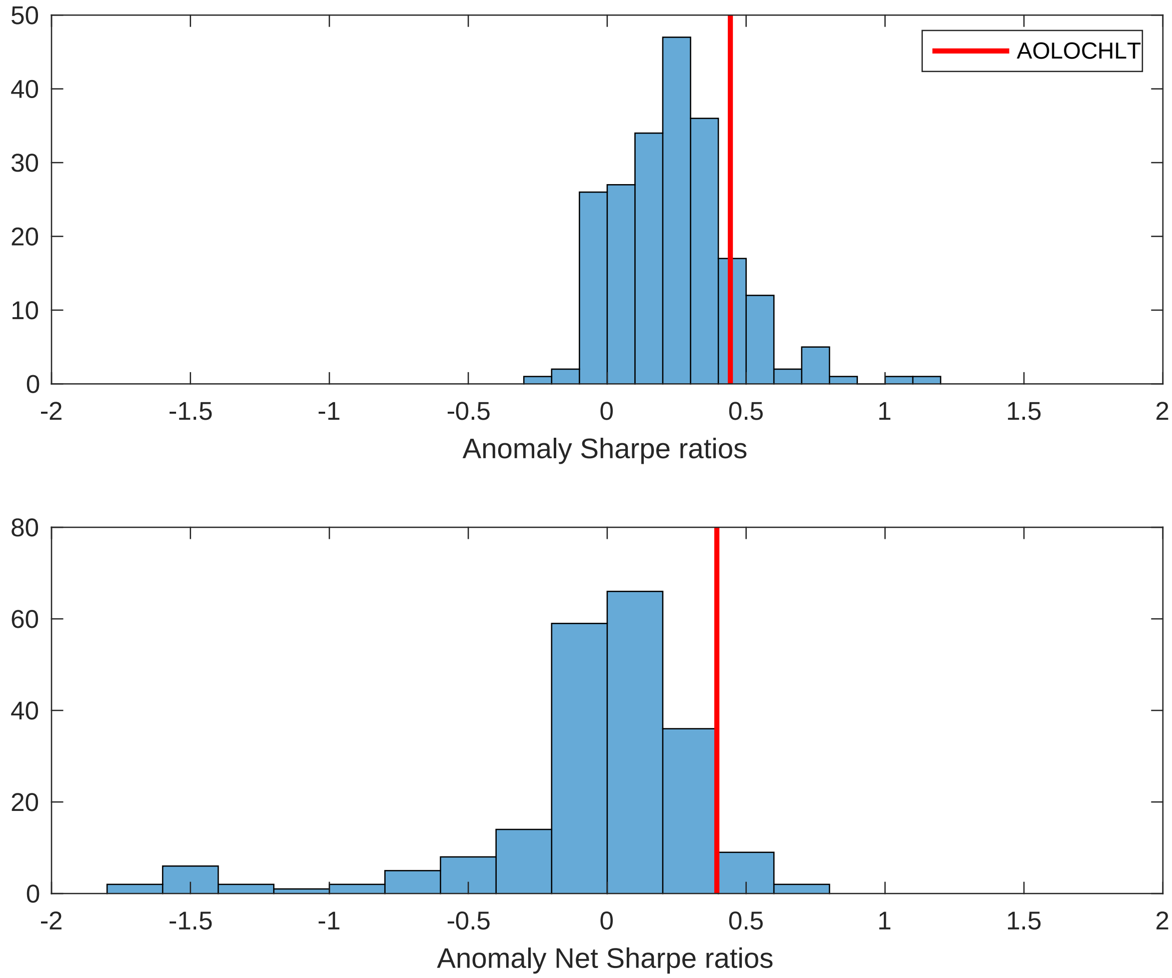


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BSV with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

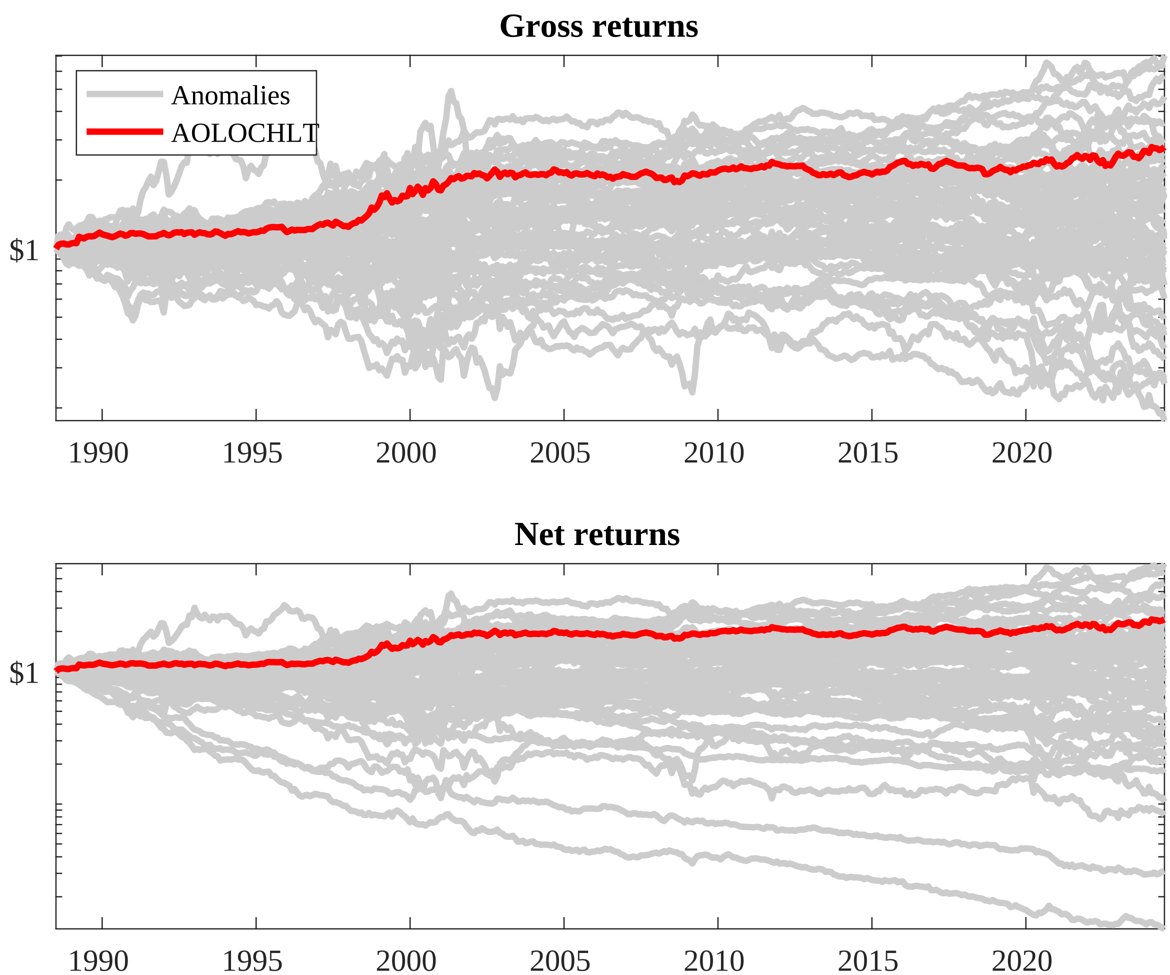
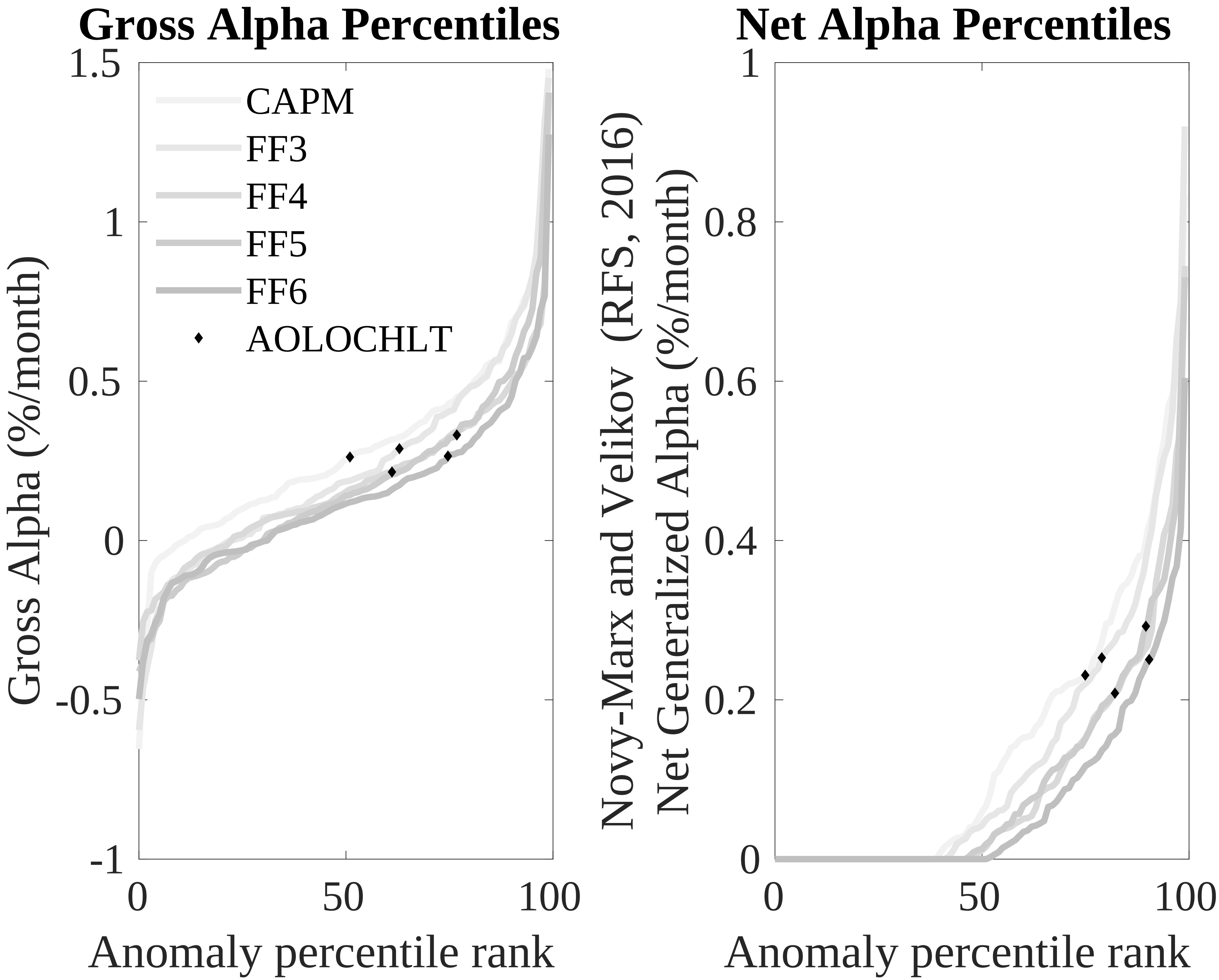


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BSV trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
AOLOCHLT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

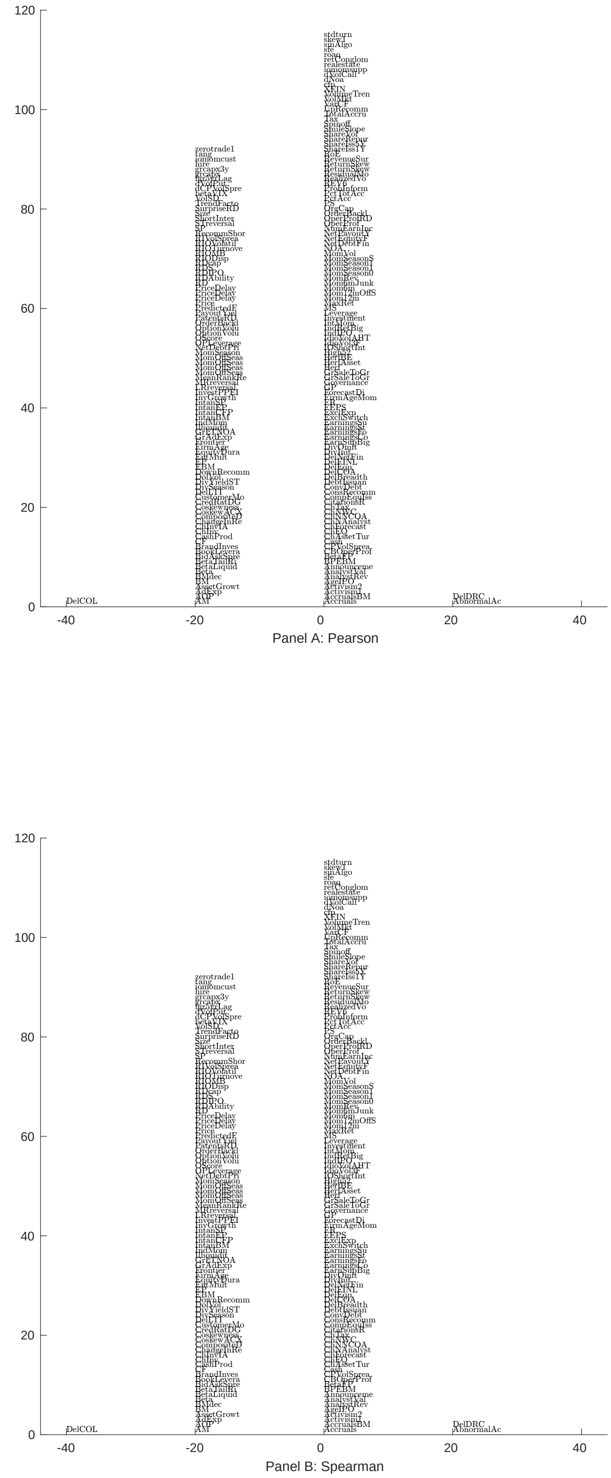


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with BSV. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

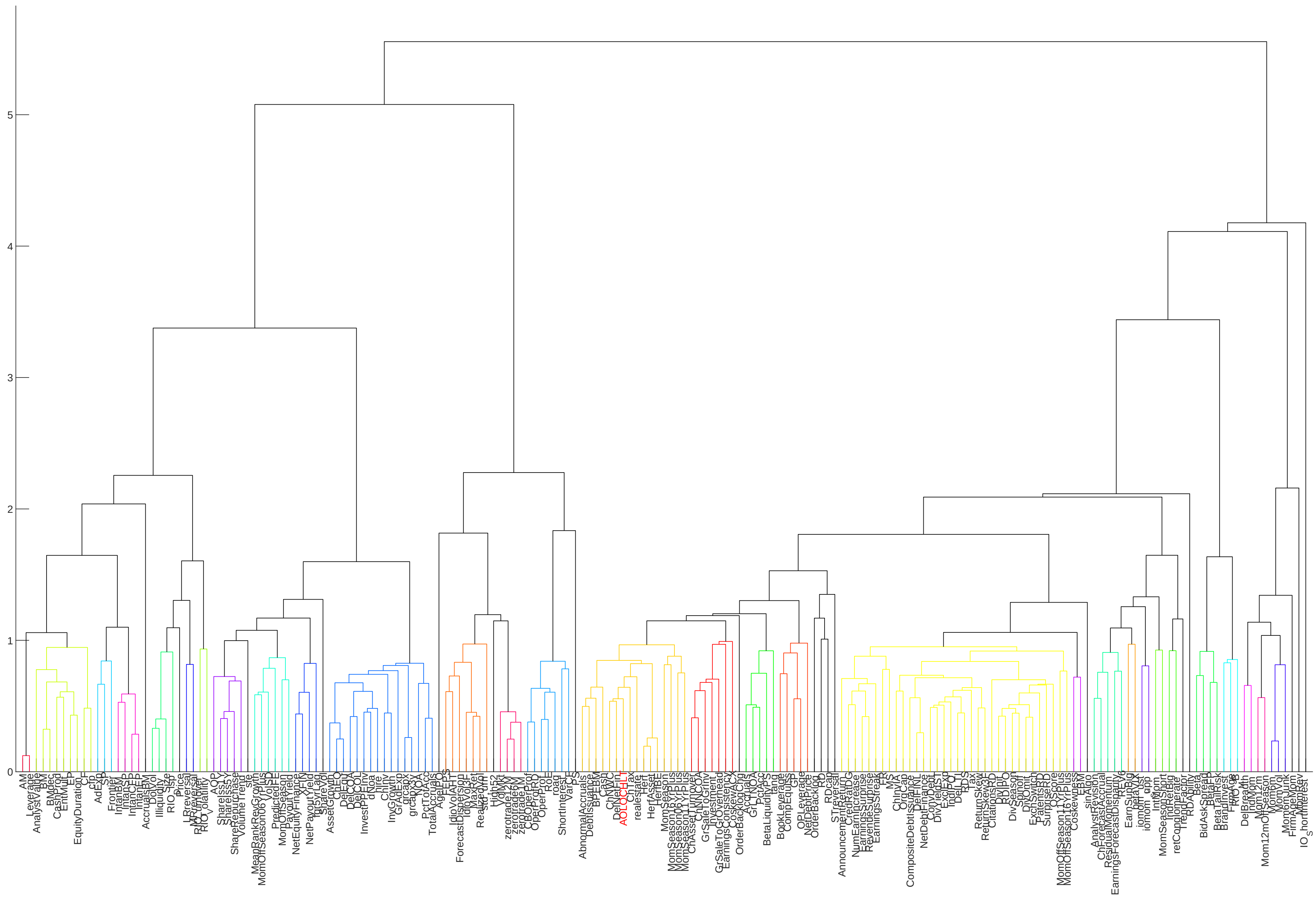


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

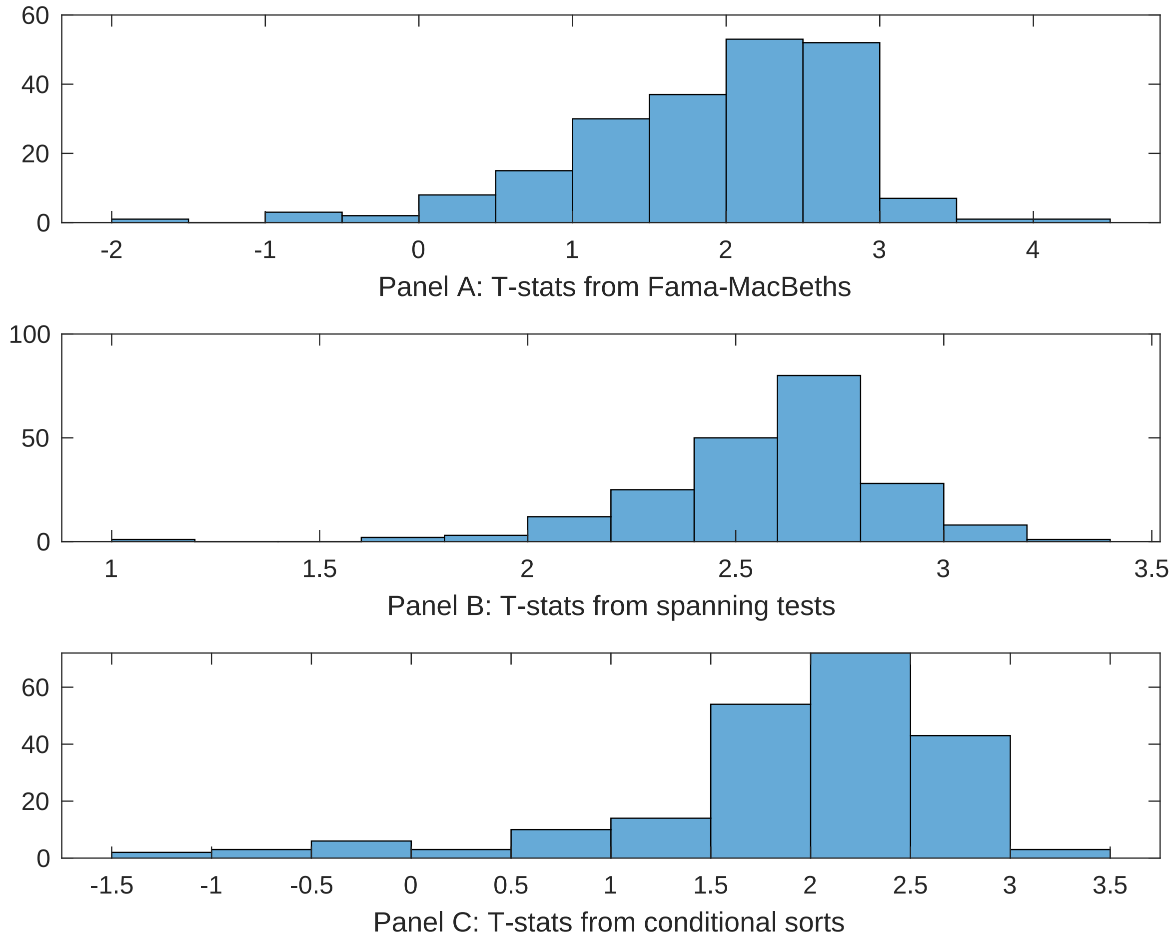


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on BSV. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BSV} BSV_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11	0.11	0.11	0.94	0.11	0.11	0.12
	[3.99]	[3.58]	[3.88]	[3.32]	[3.88]	[3.91]	[4.22]
BSV	0.64	0.51	0.40	0.52	0.45	0.39	0.71
	[3.11]	[2.57]	[1.89]	[1.75]	[2.10]	[2.05]	[2.43]
Anomaly 1	0.21						0.24
	[4.69]						[3.38]
Anomaly 2		0.14					0.19
		[1.45]					[1.72]
Anomaly 3			0.51				0.76
			[1.90]				[2.46]
Anomaly 4				0.56			-0.14
				[0.08]			[-0.21]
Anomaly 5					0.56		-0.49
					[0.19]		[-1.35]
Anomaly 6						0.14	0.14
						[5.62]	[4.73]
# months	432	432	432	427	432	432	427
$\bar{R}^2$(%)	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BSV trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BSV} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.20	0.24	0.28	0.28	0.22	0.27	0.19
	[2.09]	[2.49]	[2.91]	[2.83]	[2.24]	[2.83]	[1.99]
Anomaly 1	-22.95						-12.80
	[-5.21]						[-2.50]
Anomaly 2		15.56					6.82
		[4.46]					[1.74]
Anomaly 3			-15.31				-5.20
			[-3.46]				[-1.07]
Anomaly 4				-2.61			0.58
				[-0.88]			[0.20]
Anomaly 5					15.74		10.15
					[3.52]		[2.26]
Anomaly 6						17.60	12.09
						[4.57]	[3.06]
mkt	0.29	1.02	0.09	-0.26	-1.90	-1.93	-1.90
	[0.12]	[0.43]	[0.04]	[-0.11]	[-0.79]	[-0.81]	[-0.79]
smb	-10.32	-11.54	-9.35	-7.09	-9.22	-8.26	-12.30
	[-3.05]	[-3.33]	[-2.73]	[-1.91]	[-2.69]	[-2.45]	[-3.39]
hml	8.20	8.89	-4.51	-1.94	-0.98	-0.48	10.84
	[1.77]	[1.81]	[-1.08]	[-0.44]	[-0.23]	[-0.12]	[2.04]
rmw	-1.87	1.26	-5.69	-1.85	-0.38	-2.59	-0.74
	[-0.44]	[0.29]	[-1.29]	[-0.41]	[-0.09]	[-0.61]	[-0.17]
cma	-9.12	-20.72	-10.57	-20.14	-17.61	-19.06	-7.15
	[-1.43]	[-3.45]	[-1.56]	[-3.19]	[-2.87]	[-3.17]	[-1.01]
umd	9.24	9.87	9.06	9.29	9.67	5.83	5.65
	[4.42]	[4.70]	[4.22]	[3.82]	[4.55]	[2.52]	[2.30]
# months	432	432	432	428	432	432	428
$\bar{R}^2(\%)$	18	17	15	13	15	17	22

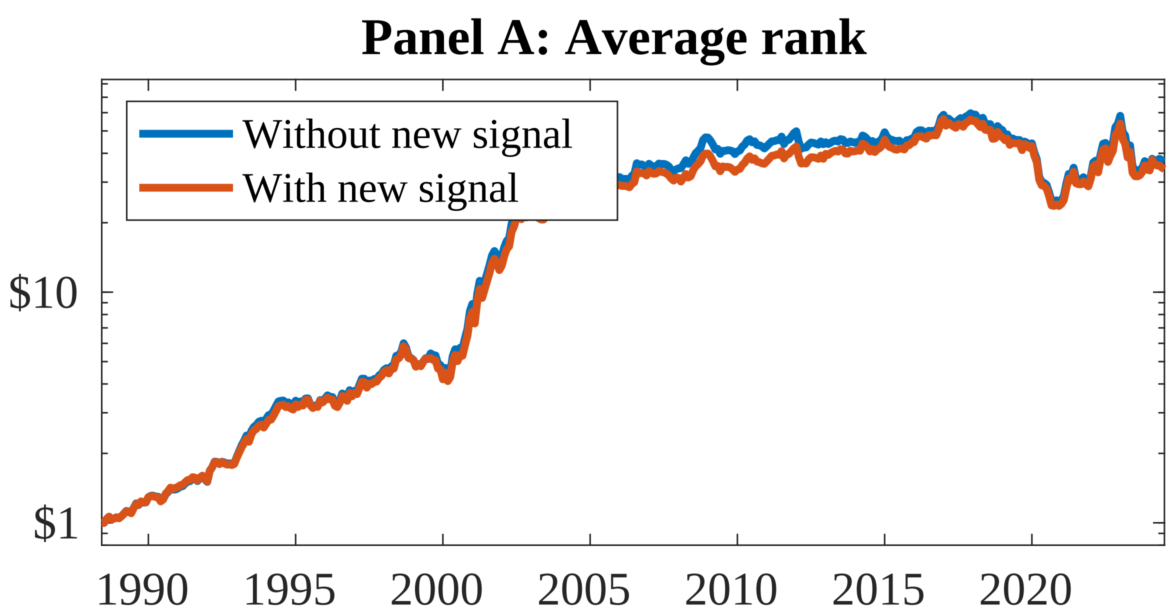


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as BSV. Panel A shows results using ”Average rank” as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Breeden, D. T. and Litzenberger, R. H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651.

Campbell, J. Y. (2018). Financial economics and asset pricing. *Review of Financial Studies*, 31(4):1253–1292.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2023). Consumption-based asset pricing: Evidence from new tests. *Journal of Financial Economics*, 148(2):245–272.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gaynor, B. and Kelly, B. (2015). Tail risk and asset prices. *Review of Financial Studies*, 28(10):2850–2883.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Journal of Finance*, 78(3):1489–1531.